1.Understanding the Sanitation Hygiene State (SHS)

The Sanitation Hygiene State (SHS) is a computed cleanliness score for a facility (like a restroom), based on real-time sensor data. Instead of a person sniffing and deciding if the room is "clean" or "dirty," a processor uses sensor readings to calculate a numeric hygiene value. This gives an objective, machine-measurable indicator of how clean the air is.

1.1 Sensing Odor-Causing Gases

The SHS system uses multiple gas sensors to capture smells or chemicals in the air that indicate dirtiness. In particular, it measures:

- Ammonia (NH_3) – often present in urine odors.
- Hydrogen sulfide (H_2S) – smells like rotten eggs.
- Volatile organic compounds (VOC) – a catch-all for other smelly chemicals (denoted as *V*).

These three readings form a multi-gas concentration vector $G(t) = [NH_3(t), H_2S(t), V(t)]$. Each sensor outputs an analog signal, which is converted into parts per million (ppm) for each gas.

1.2 Normalizing the Gas Readings

Raw sensor readings can vary widely, so the system normalizes them against known clean-air levels and bad-odor thresholds. For each gas:

- Baseline (*G_baseline*): the normal concentration in clean conditions.
- Threshold (*G_threshold*): the concentration above which we consider the odor bad.

The reading *G(t)* is scaled using:

Normalized value = (G(t) – G_baseline) / (G_threshold – G_baseline).

This means when the reading equals the clean baseline, the normalized value is 0; when it equals the threshold, the value is 1. In effect, each gas reading becomes a dimensionless deviation from clean air, on a consistent scale.

1.3 Persistence: Ignoring Short-Term Spikes

A quick, temporary whiff of odor (like someone using the restroom briefly) shouldn't immediately flag the place as dirty. To avoid false alarms, the system uses a temporal persistence factor (P). This checks if bad odors *persist* over a short time window (e.g. 60 seconds).

In practice:

- The system looks back over the last Δ seconds (say, 60 s).
- It counts how much of that time the combined odor level was above a "degradation threshold" (*T_d*).
- If the odor level is high only for a few seconds, the persistence factor stays low.
- If the odor level stays high continuously (or often) over the window, *P* increases.

This makes SHS reflect sustained odors rather than fleeting spikes. In simple terms: "*If it only smelled bad for a moment, we don't panic; if it stays bad for a minute, it matters.*"

1.4 Adjusting for Environment

Humidity and temperature can affect gas sensor readings, so the SHS calculation includes an Environmental Compensation Factor (ECF). The ECF is a linear combination of the current humidity and temperature:

$ECF(t) = k_1 * Humidity(t) + k_2 * Temperature(t)$.

Here, k_1 and k_2 are calibration constants determined in advance. This term essentially corrects the gas readings so that the hygiene score isn't thrown off by, say, a hot muggy day or a dry cold day.

1.5 Combining Factors into the SHS Score

After gathering and adjusting all pieces, the system computes the final hygiene score as a weighted sum:

$SHS(t) = w_1 * (Normalized\ NH_3) + w_2 * (Normalized\ H_2S) + w_3 * P(t) + w_4 * ECF(t)$.

- w_1, w_2: weights for the importance of ammonia and hydrogen sulfide levels.
- w_3: weight for the persistence factor (how much sustained odor influences the score).
- w_4: weight for the environmental compensation (how much we correct for humidity/temperature).

All these *w* values are pre-set in the system (memory). The result *SHS(t)* is a continuous numerical value. A higher SHS means worse air quality.

1.6 Interpreting the SHS Value

The computed SHS number is then mapped to simple categories so the system can act on it automatically. For example:

- SHS < 0.3: *"Clean State."* The air is considered essentially fresh.
- 0.3 ≤ SHS < 0.6: *"Moderate Degradation."* Some odor is present, but not severe.
- SHS ≥ 0.6: *"Sustained Degradation."* The space is noticeably smelly and may need cleaning soon.

These ranges and labels allow the system to trigger actions (like alerts or fan activation) in a clear way.

1.7 Why This Matters

By turning hygiene into a calculated number, the system achieves several benefits:

- Objective Measurement: Machines now "measure" cleanliness, replacing subjective human sniff tests.
- Consistent Monitoring: Odors are tracked over time in a precise way, making it possible to detect trends or chronic issues.
- Automated Responses: Cleaning staff or ventilation systems can be activated based on clear metrics instead of guesswork.
- Data Tracking: The hygiene score can be logged over time, providing a "sanitation history" of the facility.

In short, the SHS framework transforms "clean or dirty" from a vague judgment into a technical parameter derived from sensors. This makes sanitation monitoring more scientific, reliable, and efficient.

2.SENSOR FUSION LOGIC FOR SANITATION ASSESSMENT

Multi-Sensor Fusion Architecture

The present invention utilizes a processor-executed multi-sensor fusion architecture to determine sanitation degradation within a monitored environment. Unlike conventional systems that rely on independent threshold triggering from a single gas sensor, the disclosed system simultaneously processes multiple bio-gas sensor inputs to generate a composite sanitation assessment.

All fusion logic is executed locally at an edge processor. Only validated sanitation events or computed hygiene parameters are transmitted to a cloud-based monitoring platform. This architecture reduces bandwidth usage, improves response time, and ensures that only verified sanitation states are communicated externally.

2.1 Sensor Data Normalization

Each gas sensor generates an analog voltage proportional to the detected gas concentration. The processor converts these raw outputs into calibrated parts-per-million (ppm) values using stored calibration parameters.

Because different gases operate across different concentration ranges, direct aggregation would create scale imbalance. To ensure comparability, each gas value is normalized according to the following relation:

$$G_{norm}(t) = \frac{G(t) - G_{baseline}}{G_{threshold} - G_{baseline}}$$

Where:

- $G(t)$= real-time gas concentration
- $G_{baseline}$= reference concentration under clean conditions
- $G_{threshold}$= predefined sanitation degradation threshold

This normalization converts heterogeneous gas measurements into dimensionless values, allowing accurate mathematical fusion across sensor types.

2.2 Weighted Aggregation of Gas Signals

Following normalization, each gas component is assigned a weighting coefficient reflecting its relative sanitation significance. A composite fusion score is computed as:

$$S_{fusion}(t) = w_1 \times NH_3^{norm}(t) + w_2 \times H_2S^{norm}(t)$$

Where:

- w_1, $w_2$are weighting coefficients stored in system memory.

In certain embodiments, hydrogen sulfide may be assigned a higher weight due to its stronger association with organic decomposition events. Weighted aggregation ensures that sanitation classification is not disproportionately influenced by an isolated spike from a single gas sensor.

2.3 Cross-Correlation Analysis

To improve reliability, the system computes temporal cross-correlation between ammonia and hydrogen sulfide concentration values over a sliding time window.
Sustained sanitation degradation typically produces correlated increases in both gases. In contrast, transient disturbances—such as brief occupancy or chemical exposure—often result in weak, inconsistent, or uncorrelated patterns.
If the calculated correlation coefficient exceeds a predefined threshold, the system increases confidence that a genuine sanitation degradation event is occurring.
This correlation-based validation adds a behavioral dimension to gas measurement rather than relying solely on magnitude.

2.4 Gas Ratio Pattern Detection

The processor further computes a dynamic gas ratio:

$$R(t) = \frac{NH_3(t)}{H_2S(t)}$$

Biological waste events typically produce characteristic ratio bands within expected ranges. By contrast, exposure to cleaning chemicals or short-term occupancy may cause disproportionate ammonia increases without corresponding hydrogen sulfide levels.
Ratio-based pattern recognition therefore enables differentiation between sustained biological degradation and temporary odor disturbances, enhancing classification accuracy.

2.5 Temporal Persistence Filtering

Sanitation degradation is validated only if the composite fusion score exceeds a predefined degradation threshold for a minimum persistence duration.
If:

$$S_{fusion}(t) > T_d$$

for a duration exceeding T_{min}, the event is classified as sustained degradation.
Short-duration spikes that do not exceed the persistence window are automatically rejected. This temporal filtering prevents false alerts triggered by brief or incidental odor fluctuations.

2.6 False Positive Suppression Mechanisms

The system incorporates multiple validation layers to reduce false positives, including:

- Rejection of single-gas deviation without cross-gas confirmation
- Filtering of spikes shorter than a predefined duration threshold
- Correlation validation between gas vectors
- Environmental compensation using humidity and temperature factors

These mechanisms collectively improve signal stability, reduce unnecessary maintenance alerts, and enhance long-term operational reliability.

2.7 Composite Hygiene Signal Output

The final sanitation assessment signal is computed as:

$$SHS_{raw}(t) = w_1 \times NH_3^{norm}(t) + w_2 \times H_2S^{norm}(t) + w_3 \times Correlation(t) + w_4 \times Persistence(t)$$

Where:

- $Correlation(t)$represents the cross-gas correlation coefficient
- $Persistence(t)$represents the time-above-threshold factor
- $w_1, w_2, w_3, w_4$are weighting coefficients stored in system memory

The resulting $SHS_{raw}(t)$is a continuous numerical sanitation state parameter used for hygiene classification and degradation detection.

2.8 Technical Superiority Over Single-Threshold Systems

Conventional single-threshold systems trigger alerts when a single gas exceeds a predefined concentration level. Such systems are unable to distinguish between sustained degradation events and short-term odor spikes, leading to excessive false alarms and inefficient cleaning dispatch.

By contrast, the present multi-sensor fusion system integrates:

- Normalization
- Weighted aggregation
- Cross-correlation analysis
- Ratio pattern detection
- Temporal persistence modeling
- Environmental compensation

This multi-dimensional sanitation vector approach produces a stable and reliable sanitation degradation signal.

As a result, the system achieves:

- Improved detection precision
- Reduced false positive rates
- Optimized maintenance resource allocation
- Enhanced operational efficiency

Through structured sensor fusion and temporal modeling, sanitation assessment is transformed from a threshold-trigger mechanism into a robust computational classification process.

3.EDGE INTELLIGENCE FOR REAL-TIME SANITATION PROCESSING

3.1 Functional Edge Processing Architecture

The present invention incorporates an embedded edge-processing module implemented on a microcontroller-based hardware platform. The edge processor performs real-time sanitation assessment computations locally, prior to any cloud communication.

The use of edge processing is not a mere architectural distribution of software components. Rather, it is functionally required to achieve measurable technical improvements, including reduced latency, optimized bandwidth utilization, and improved operational reliability.

The edge processor is operatively coupled to:

- Gas sensors
- Environmental sensors
- Communication modules
- Local output indicators

The processor executes sanitation classification logic directly on sensor-derived physical measurements.

3.2 Technical Requirement for Edge Processing

Edge-level computation is technically necessary due to operational constraints inherent in sanitation monitoring environments:

1. Continuous sensor sampling produces high-frequency raw data streams.
2. Sanitation degradation detection requires immediate computational validation.
3. Public facilities may experience unstable or intermittent network connectivity.
4. Transmission of unfiltered raw sensor data to a remote server would substantially increase bandwidth consumption and introduce processing latency.

Accordingly, sanitation decision logic is executed locally at the hardware level before data transmission occurs. This ensures deterministic behavior and consistent performance irrespective of network conditions.

3.3 Pre-Transmission Processing Operations

Prior to any cloud communication, the edge processor performs a sequence of real-time computational operations, including:

- Analog-to-digital conversion of sensor signals
- Calibration and normalization of gas concentration values
- Multi-sensor fusion processing
- Cross-correlation analysis between gas vectors

- Temporal persistence validation
- Environmental compensation adjustment
- Sanitation Hygiene State (SHS) computation
- Classification into Clean, Moderate Degradation, or Sustained Degradation states
- Event compression and structured data packet generation

Only validated sanitation state transitions or confirmed degradation events are transmitted to the cloud platform.
This local validation step ensures that raw sensor fluctuations are not directly propagated to remote systems.

3.4 Data Reduction and Bandwidth Optimization

Gas and environmental sensors may sample at predefined intervals (for example, every 5 seconds). Continuous transmission of such raw data streams would result in substantial bandwidth usage across distributed installations.
By performing sanitation classification locally, the system transmits only:

- Sanitized event summaries
- Degradation start and end timestamps
- Computed hygiene state values
- Aggregated trend data, where required

This approach significantly reduces data volume compared to continuous raw streaming.
Edge-level preprocessing therefore achieves measurable bandwidth reduction and network load optimization, enabling scalable deployment across multiple facilities.

3.5 Latency Reduction and Immediate Response Capability

Effective sanitation monitoring requires near real-time detection of degradation events.
If sanitation decision logic were executed solely in a cloud environment, network transmission delays could introduce unacceptable response latency.
By computing sanitation classification locally:

- Degradation detection occurs immediately upon threshold validation.
- Alerts can be triggered without waiting for server acknowledgment.
- Local indicators, such as buzzers, warning lights, or display panels, may be activated instantly.

This ensures low-latency response and immediate operational action.

3.6 Reliability During Network Disruption

Public sanitation facilities often operate in environments where network connectivity may be unstable or temporarily unavailable.
The edge processor maintains autonomous operational capability independent of cloud connectivity. If communication is disrupted:

- Sanitation classification continues uninterrupted.
- Degradation events are stored in local memory.
- Logged data is transmitted when connectivity is restored.

This autonomous capability enhances system robustness and ensures continuous sanitation monitoring regardless of external network conditions.

3.7 Alert Flooding Prevention

Temporary odor spikes and environmental fluctuations are common in high-traffic sanitation facilities.
If raw sensor data were transmitted directly to a remote server without local validation, excessive false alerts could be generated.
Edge-level classification prevents alert flooding by:

- Filtering short-duration spikes
- Rejecting single-gas deviations without cross-gas confirmation
- Validating sustained degradation through temporal persistence modeling
- Transmitting only confirmed sanitation events

This controlled alert mechanism ensures that maintenance notifications are meaningful and actionable.

3.8 Hardware-Integrated Processing Implementation

The edge intelligence module is implemented within a microcontroller comprising:

- Onboard memory storing sanitation models and weighting coefficients
- Sensor interface circuitry

- Wireless communication module
- Real-time clock module

The sanitation assessment computations are executed directly by the hardware processor and depend on physical sensor inputs representing measurable environmental conditions.
The system therefore constitutes a hardware-integrated technical solution rather than an abstract software algorithm executed independently of physical components.

3.9 Technical Effect Achieved

The edge intelligence module provides demonstrable technical advantages, including:

- Reduced data transmission bandwidth
- Reduced network latency
- Increased system reliability
- Autonomous degradation detection
- Controlled and validated alert generation
- Improved energy efficiency

The edge processing architecture is therefore functionally necessary to achieve measurable technical performance improvements. It is not merely a relocation of algorithmic logic but an essential component enabling efficient, real-time sanitation monitoring within hardware-constrained environments.

4. AI MODEL - OPERATIONAL DISCLOSURE

TO BE CONT…

5.UNCLEAN-PERIOD DETECTION MODULE:

5.1 Detection Mechanism:

The system defines an **Unclean Period** as a measurable sanitation degradation interval determined by computed Sanitation Hygiene State (SHS).

5.1.1 Start of Unclean Period:

An Unclean Period is initiated when:

- The computed SHS value is **greater than or equal to 0.6.**
- The SHS remains above this threshold continuously for a minimum duration of **60 seconds.**

This persistence-based condition prevents false triggering due to transient odor spikes.
Mathematically:
If SHS(t)>= 0.6 for t >=60 s:
Degradation Start Time = T_degradation

5.1.2 End of Unclean Period:

The Unclean Period terminates when:

- SHS falls below **0.3**
- The SHS remains below 0.3 continuously for at least **90 seconds**.

Mathematically:
If SHS(t) < 0.3 for t >= 90 s
Restoration Time = T_restore
The use of different start and end thresholds introduces hysteresis, thereby improving detection stability.

5.2 Duration Measurement:

The duration of the unclean period is computed as :

$$T_unclean = T_restore - T_degradation$$

Where:

T_degradation = timestamp of sustained degradation detection
T_restore = timestamp of sustained hygiene restoration

The system logs this duration in non-volatile memory and transmits it to the cloud database.

5.3 Technical Effect:

The Unclean-Period Detection Module produces a measurable operational parameter, enabling:

- Quantifiable sanitation degradation duration
- Cleaning response efficiency assessment
- Performance-based maintenance tracking
- Objective Service-Level Agreement (SLA) compliance monitoring

The generation of a duration-based sanitation degradation metric constitutes a technical effect beyond mere threshold alerting.

6. ALERT VALIDATION AND USER FEEDBACK PROCESSING:

6.1 Sensor-Verified Complaint Validation:

When a user submits a sanitation complaint via the interface, the system performs real-time cross-verification with sensor-derived SHS values.
A complaint is classified as **Valid** only if:

- SHS > 0.5, and
- Persistence of this condition exceeds 30 seconds.

Condition:

If (SHS > 0.5) AND (Persistence > 30 s)
    Valid Complaint

Otherwise, the complaint is classified as a **Transient Anomaly**.

6.2 Technical Contribution:
This mechanism ensures:

- Elimination of false or malicious complaints
- Reduction in unnecessary dispatch events
- Conversion of subjective feedback into sensor-validated technical data

The validation logic operates at system level and is integrated with the edge computation module.

7. HARDWARE ARCHITECTURE (HARDWARE DEPENDENCE)

The sanitation assessment system comprises:

- ESP32 microcontroller configured as edge processor
- Onboard flash memory storing hygiene model parameters
- MQ-135 gas sensor interface (Ammonia detection)
- MQ-136 gas sensor interface (Hydrogen Sulfide detection)
- Wi-Fi communication module
- Optional GSM communication module
- 5V regulated DC power supply
- Cloud dashboard server for monitoring and analytics

The system requires physical sensor interfaces and hardware-based computation, thereby establishing hardware dependence.

8. DETAILED EMBODIMENT (SECTION 10(4) COMPLIANCE)

8.1 Operational Workflow:

Step 1 - Sensing

Gas sensors acquire NH_3 and H_2S concentrations at a sampling interval of 5 seconds.

Step 2 - Normalization

Raw ppm values are converted into normalized hygiene deviation vectors using baseline calibration parameters stored in memory.

Step 3 - Sensor Fusion

Edge processor performs:

- Rolling window averaging
- Gas ratio computation
- Weighted SHS calculation

Step 4 - Hygiene Classification

Computed SHS is compared with predefined thresholds to classify hygiene state as:

- Clean
- Moderate
- Degraded

Step 5 - Degradation Detection

Persistence logic determines whether the state qualifies as sustained degradation.

Step 6 - Alert Transmission

Upon validation, a structured cleaning alert is transmitted to the cloud dashboard.

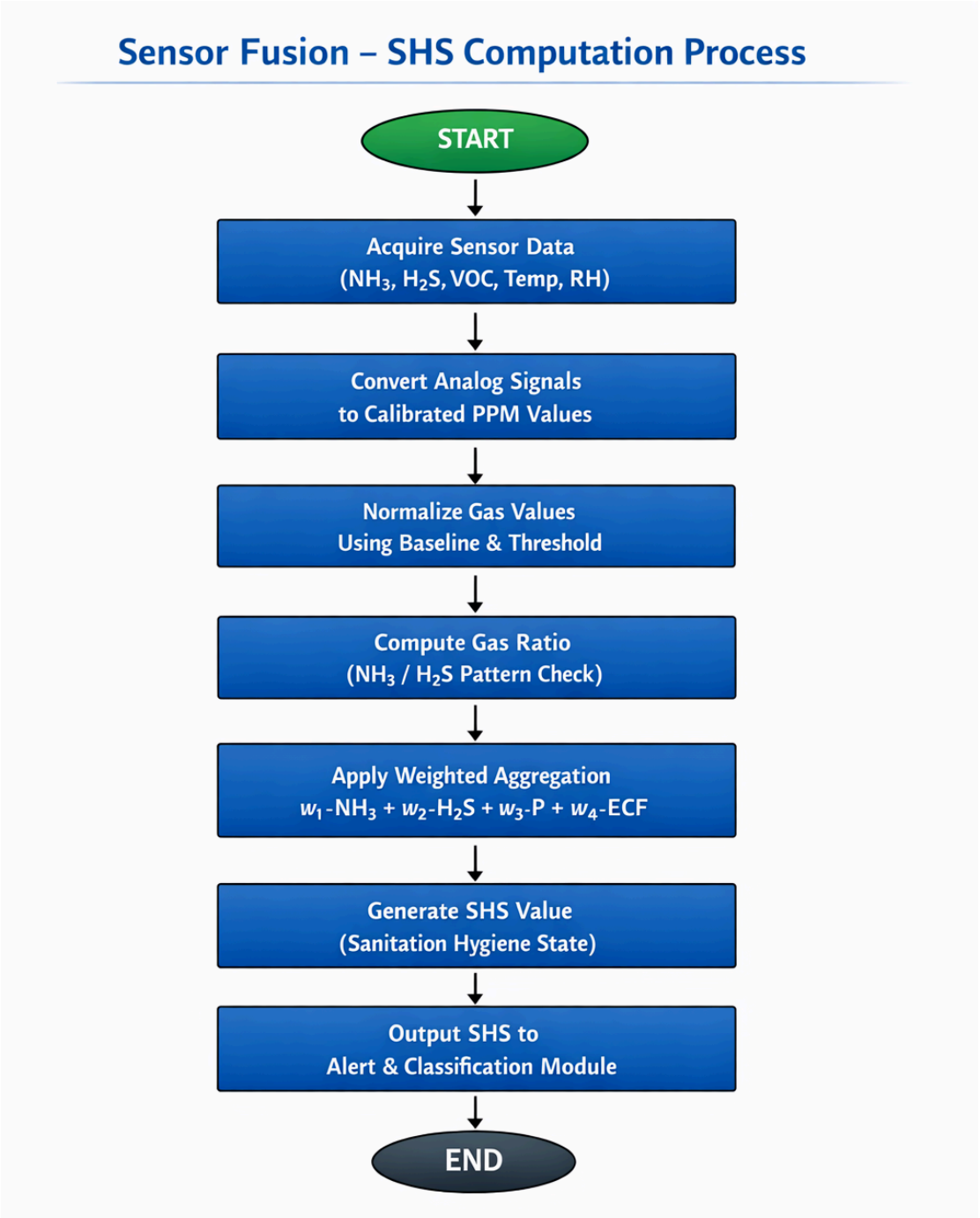
Step 7 - Logging

The calculated Unclean Period duration is stored locally and synchronized with the cloud database.

9. FLOWCHART DESCRIPTIONS

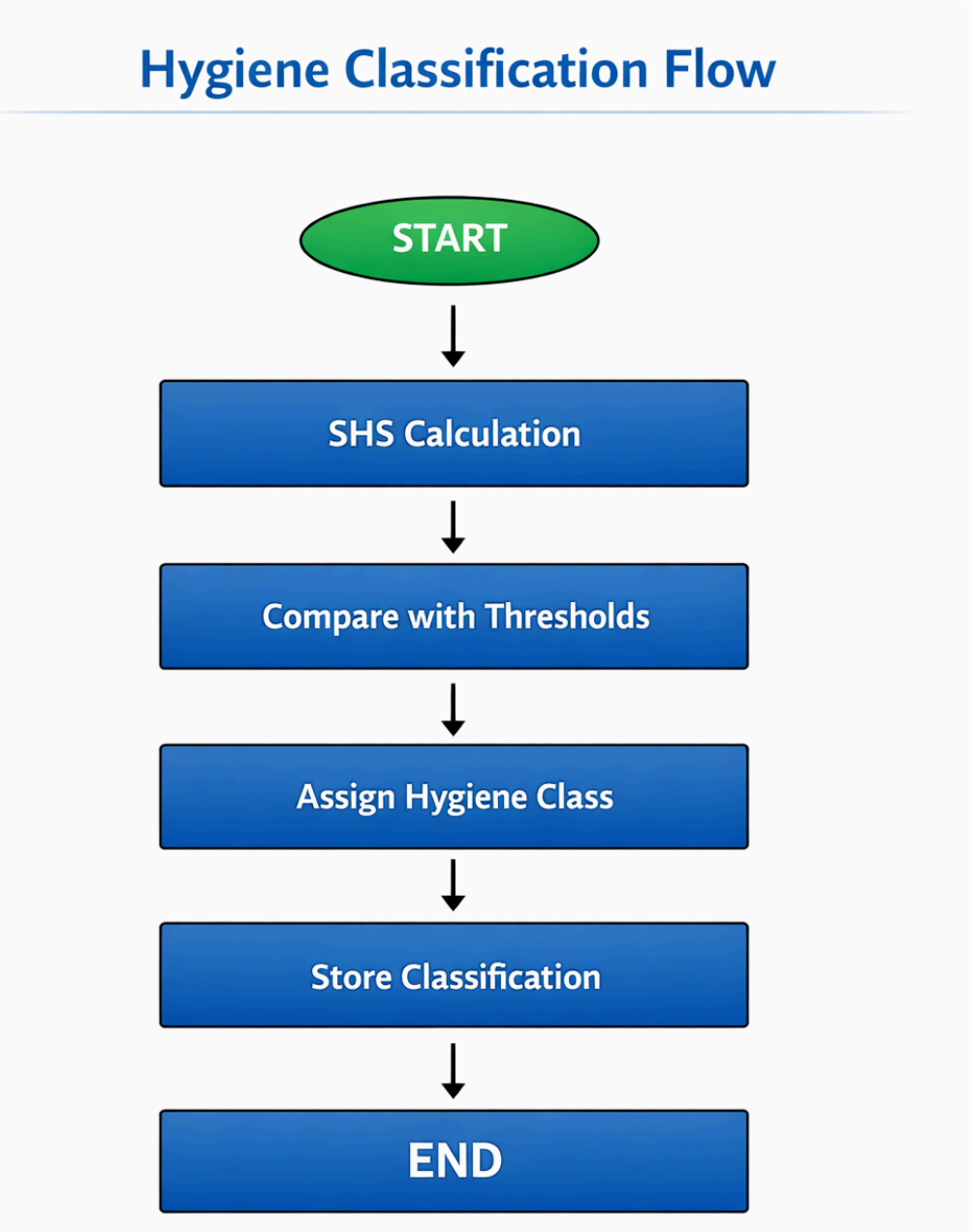
9.1 Sensor Fusion Flow:

Acquire Sensor Data
Normalize values
Compute gas ratio
Apply weigjted sum
Generate shs output



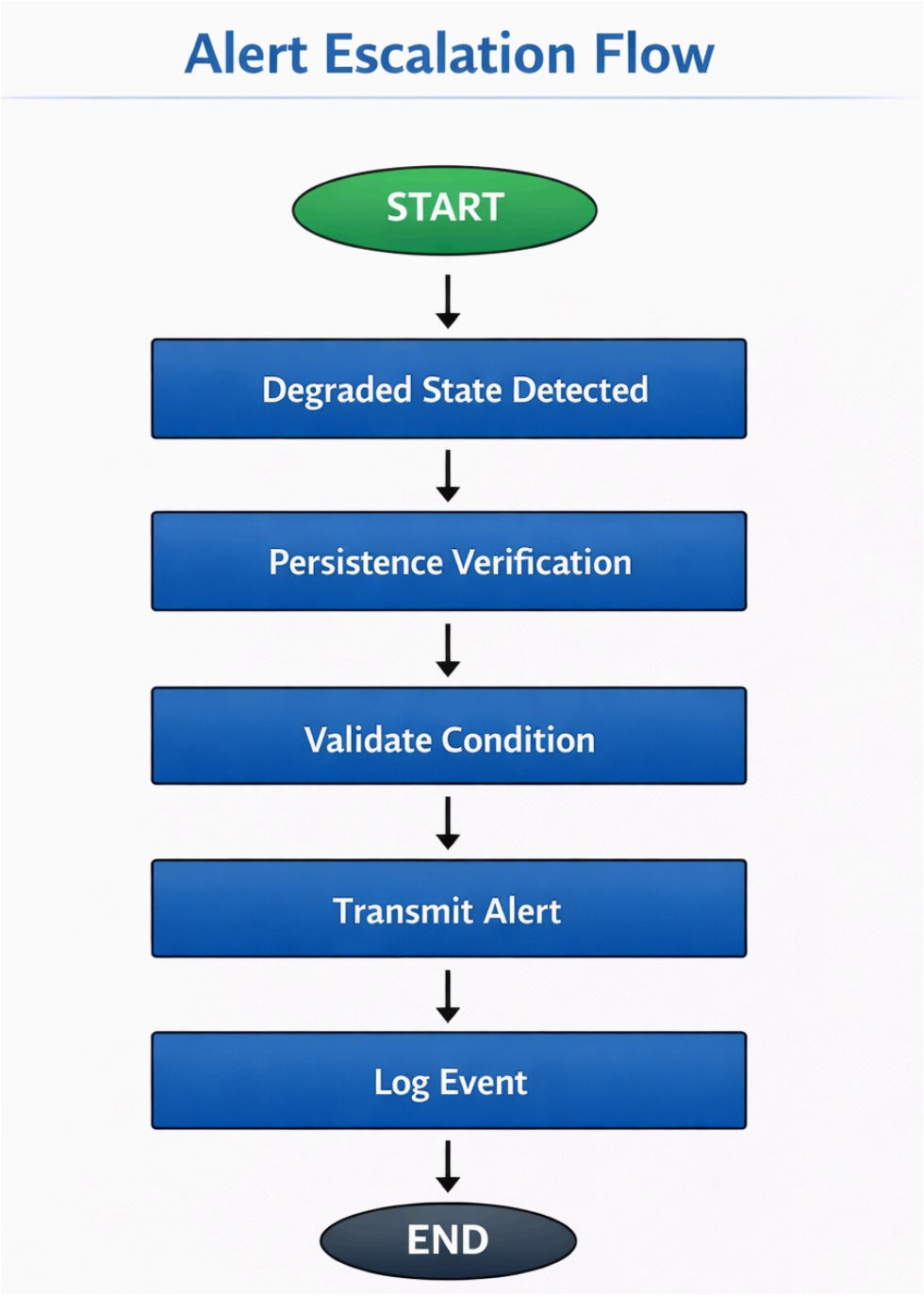
9.2 Hygiene Classification Flow:

SHS Calculation
Compare with Thresholds
Assign Hygiene Class
Store Classification



9.3 Alert Escalation Flow

Degraded State Detected
Persistence Verification
Validate Condition
Transmit Alert
Log Event



10. BEST MODE DISCLOSURE

The preferred embodiment includes:

Preferred Gas Sensors
- MQ-135 for Ammonia detection
- MQ-136 for Hydrogen Sulfide detection

Preferred Sampling Interval
- 5 seconds per acquisition cycle

Preferred Persistence Window
- 60 seconds for degradation confirmation

Preferred Communication Mode
- Wi-Fi as primary communication medium
- GSM as fallback redundancy channel